

No more PD/LGD/EAD

Simple Unified Approach to Loans and Receivables for Regulatory Capital Requirements and Carrying Values

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Abstract

As regards credit risk, the accounting and prudential regulatory worlds ask two questions: (1) What is a loan worth? and (2) How much could that value fall in extreme circumstances? The first determines the carrying value of the asset and hence balance sheet equity; the second determines capital requirements by assessing possible reductions in that carrying value and hence reduction in equity. This matters in part because bank debt covenants reference accounting equity — breach them and creditors can liquidate, creating systemic risk. This is where the Basel capital accords come in as this is a risk that may not be taken into account by the banks, so they must be regulated. In practice, in attempting to answer these two questions, the current frameworks of Basel and IFRS obscure these simple questions with circular definitions and artificial decompositions. We propose: value loans at expected cash flow NPV for accounting, use the 99.9th percentile of NPV decline for capital. One Monte Carlo model, two outputs, no gaming.

1 The Only Two Questions That Matter

Bank regulation exists in part because balance sheet insolvency triggers creditor liquidation rights through covenant breaches, creating systemic externalities. This creates two fundamental questions:

Question 1 (Accounting): What is the loan worth today?

Question 2 (Regulation): How much could that value fall in extreme circumstances?

That's it. Everything else is detail.

Question 1 determines balance sheet equity. Question 2 determines how much contribution to a capital buffer prevents covenant breaches in tail events (yes, even this assumes perfect correlation with other sources of equity reduction, one of the reasons stress testing is a superior way to determine capital requirements). Bank debt covenants reference accounting measures—when breached, creditors can demand repayment or liquidate collateral, potentially triggering systemic crisis.

Current frameworks fail both questions. IFRS 9 defines expected losses as "the losses you expect"—pure circularity on Question 1. Basel's PD/LGD decomposition creates gaming opportunities whilst obscuring Question 2. Meanwhile, thousands of PhDs optimise logistic regressions instead of modelling actual economics.

2 Question 1: What Is It Worth?

Generally, loans and receivables are carried at the nominal amount due. Interest payments are taken into income when they are earned. more or less. But then there is the "you might not actually get all of that money back" or "actually you are going to get nothing back" aspects - provisions and write offs. Most of the problems begin here, but already the concept of "interest income" has muddled the waters in my opinion.

2.1 Why IFRS 9 Fails

IFRS 9 asks banks to estimate "expected credit losses" without defining what constitutes a loss... in fact the standard is circular: "EAD" is the maximum *loss* and "LGD" the *loss* we expect given some event "default"...

IFRS 9's EAD approach isn't just flawed - it's intellectually bankrupt. Expected credit losses are the "losses you expect." The very notion of EAD requires us to forecast future assets - but this is the role of financial planning and regulatory capital calculations, And this is just as true whether 2008 happened or not.

As for all the buckets and "significant change in credit risk" - give me a break!

Most of this is a capitulation to "conservatism" (never has been the right word for crudely overestimating things), a nod to regulation and a severe departure from true and fair. A departure actually from what makes accounting fascinating.

2.2 The Answer: Expected Cash Flows

A loan is worth the present value of expected cash flows:

$$\text{Loan Value} = \sum_{t=1}^T \frac{E[\text{Cash Flow}_t]}{(1+r)^t} \quad (1)$$

This isn't controversial—it's how we value everything else. Bonds, equities, derivatives, real estate—all valued at expected cash flow NPV. Only loans get special treatment with "amortised cost less expected credit losses," creating the circular definition problem.

3 Question 2: How Much Could the Value Fall?

3.1 The Answer: Distribution Tails

Run Monte Carlo simulations with systematic factors affecting payment probabilities:

$$P_{i,t} = f(\mathbf{S}_t, I_{i,t}, \mathbf{X}_{i,t}) \quad (2)$$

where systematic factors $\mathbf{S}_t$ (GDP, rates, unemployment) drive correlated defaults, idiosyncratic factors $I_{i,t}$ capture borrower-specific risks, and $\mathbf{X}_{i,t}$ includes observable characteristics.

Generate thousands of scenarios. The 99.9th percentile of value decline is your capital requirement:

$$\text{Capital} = \text{VaR}_{99.9\%}[\text{NPV Decline over 12 months}] \quad (3)$$

3.2 Why PD/LGD Fails

The PD/LGD first suffers from the same circularity of IFRS9 - really we should be asking "what happens to IFRS9 losses in stress" but IFRS9 is circular. Indeed, IFRS9 has us double count requires capital -once via IFRS9 and once by regulatory calculations! Again, let's rub it in "EL

= "maximum value of **L**" × percent of that you expect to lose...circular.

And of course "PD" harks back to IAS9 really looking, for a discrete event. The looseness in definition of PD also for gaming (both EL and RWA (yes, EL too)) will change if we redefine PD; as a special example, and I can reduce RWA by assigning exposure to increasingly narrow probability intervals.

Direct NPV simulation eliminates this—you can't game cash flows that either arrive or don't.

4 One Framework, Both Answers

One Monte Carlo engine answers both questions.

For Accounting (Question 1):

$$\text{Loan Value} = E[\text{NPV}] \text{ (expected value of distribution)} \quad (4)$$

For Capital (Question 2):

$$\text{Capital Buffer} = \text{VaR}_{99.9\%}[\text{NPV Decline}] \text{ (tail of distribution)} \quad (5)$$

One model. Two outputs. No parallel frameworks, no reconciliation, no gaming.

4.1 This Already Works: CVA

Banks already do this for derivatives through Credit Valuation Adjustment (CVA):

- Value derivatives at expected cash flow NPV
- Use Monte Carlo for capital requirements
- Accepted by regulators globally for trillions in exposures...

If NPV with Monte Carlo works for complex derivatives, rejecting it for simple loans is intellectual inconsistency.

5 Why This Matters

5.1 Gaming Eliminated

You can't manipulate PD/LGD splits that don't exist; can't exploit risk grade granularity; can't hide behind definitional ambiguity; either you can model cash flows or you can't....

5.2 Talent Unleashed

Thousands of PhDs currently optimising logistic regressions for regulatory arbitrage could instead build:

- Sophisticated systematic factor models
- Complex correlation structures
- Advanced econometric frameworks
- Genuine economic relationships...

5.3 Simplicity Achieved

Banks: One model for business and regulation, Regulators: Clear measurement of covenant breach risk, Auditors: Observable cash flows, not subjective "expectations," and Investors: Transparent asset values and risk buffers...

6 Implementation

Banks with good data model cash flows directly. What is proposed here is little more than current LGD approaches. Banks without data use standardised approaches. Natural differentiation based on genuine capability, not regulatory optimisation skill.

Climate risk? Just another systematic factor. Concentration risk? Emerges from factor correlations. Stress testing? Shock the systematic factors. Everything flows from the same framework.

7 Conclusion

As regards credit risky assets, the accounting and prudential regulatory worlds ask two questions: What's it worth? How much could that fall? Current frameworks obscure both with circular definitions and artificial decompositions that enable gaming whilst wasting intellectual talent.

NPV accounting answers Question 1: a loan's worth its expected cash flow present value. Monte Carlo simulation answers Question 2: capital requirements equal the 99.9th percentile of value decline. We need one framework, both answers, no gaming.

The framework exists — CVA proves it works. The talent to compute this exists — trapped optimising regulatory artifacts. Implementation requires only recognising that loans, like every other financial asset, are worth what you expect to collect.

So, if we have two questions. Let's just answer them coherently?.